

MODELING ELECTORAL BEHAVIOUR WITH PENNSYLVANIA ELECTORAL DATA

SMAC TALK

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SPATIAL VOTING MODEL

- ▶ While electoral behaviour is deeply discussed, one of the most successful mathematical models of voting is the spatial voting model
- ▶ Its basic assumption is that you can represent both voters and candidates in a ideological space, where you can define a distance
- ▶ Then, each voter votes for the candidate which is closer
- ▶ This choice structure can be adequately described by the following utility function:

$$u_i(\mathbf{x}_D) = u_i(\mathbf{x}_i, \mathbf{x}_D) + \epsilon_{iD} = -(\mathbf{x}_i - \mathbf{x}_D)' \mathbf{W}_i (\mathbf{x}_i - \mathbf{x}_D) + \epsilon_{iD}$$

DEVELOPING ON SPATIAL VOTING MODEL

- ▶ Following Merlo and Paula (2017) we know that if we impose the same weighting matrix to each voter, then we are able to identify the distribution of voters' positions
- ▶ But that constraint seems too strict! We recover variability adopting a structural demand model

$$\begin{aligned} u_i(\mathbf{x}_D) &= \sum_{t=1}^K \beta_{it} x_{Dt} + \epsilon_{iD} \\ &= \sum_{t=1}^K (\beta_t + \sum_{s=1}^K x_{is} w_{ts} + \sum_{m=1}^M c_{im} \tilde{\beta}_{tm})(x_{Dt}) + \epsilon_{iD} \\ &= \mathbf{x}_D' \beta_0 + \mathbf{x}_D' W \mathbf{x}_i + \mathbf{x}_D' \widetilde{\mathbf{B}} \mathbf{c}_i + \epsilon_{iD} \\ &= \mathbf{x}_D' (\beta_0 + W \mathbf{x}_i + \widetilde{\mathbf{B}} \mathbf{c}_i) + \epsilon_{iD} \end{aligned}$$

OUR MODEL FOR TWO-PARTY ELECTIONS

- ▶ Following Merlo and Paula (2017) we know that if we impose the same weighting matrix to each voter, then we are able to identify the distribution of voters' positions
- ▶ But that constraint seems too strict! We recover variability adopting a structural demand model
- ▶ $u_i(\mathbf{x}_D) - u_i(\mathbf{x}_R) = (\mathbf{x}_D - \mathbf{x}_R)'(\beta_0 + \mathbf{W}\mathbf{x}_i + \widetilde{\mathbf{B}}\mathbf{c}_i) + (\epsilon_{iD} - \epsilon_{iR})$
- ▶ $\Pr(y_{iD} = 1 \mid \mathbf{v}_i) = \Pr((\epsilon_{iD} - \epsilon_{iR}) \leq (\mathbf{x}_D - \mathbf{x}_R)'\mathbf{B}\mathbf{v}_i)$

ESTIMATION

- ▶ If we assume $(\epsilon_{iD} - \epsilon_{iR})$ to follow a Uniform distribution over an adequately large support $[-a, a]$, we get the following expected share:

- ▶
$$\mathbb{E}[s_D] = \frac{1}{2} + \frac{(\mathbf{x}_D - \mathbf{x}_R)'}{2a} \mathbf{B} \sum_{i=1}^N \frac{\mathbf{c}_i}{N}$$

- ▶ At a precinct level, we have:

- ▶
$$\mathbb{E}[s_{Dg} - \frac{1}{2}] = (\mathbf{x}_D - \mathbf{x}_R)' \left(\frac{\beta_0}{a} + \mathbf{W} \frac{\bar{x}_g}{a} + \sum_{m=1}^M \frac{\tilde{\beta}_{1m}}{a} \bar{c}_{mg} \right)$$

CONSTRUCTION OF THE DATASET

- ▶ From ALARM dataset (Kenny and McCartan, 2021), we had electoral results and ethnic composition at a precinct level
- ▶ From census data, we took data on educational attainment and reapportioned them at precinct level
- ▶ Finally, we looked for data on candidates' ideology: we needed a score both for winning and losing candidates, then we adopted DIME score (Bonica, 2024)

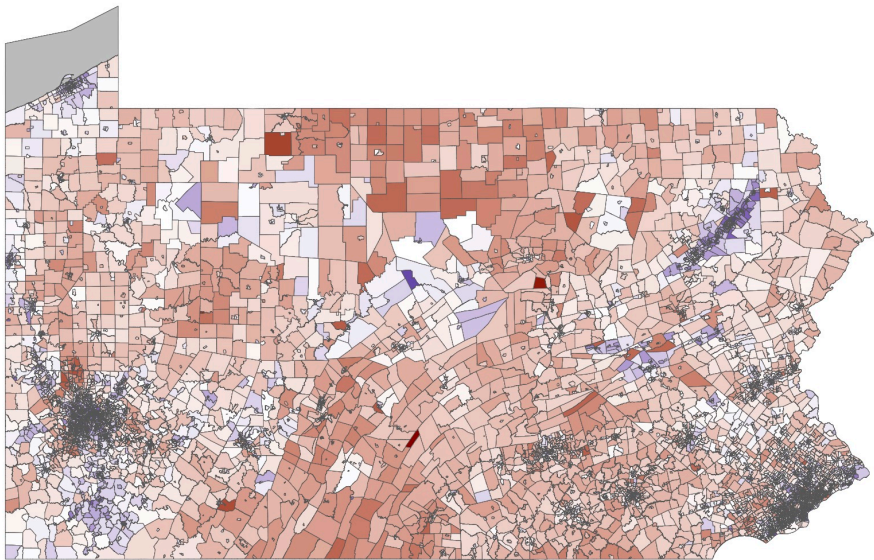
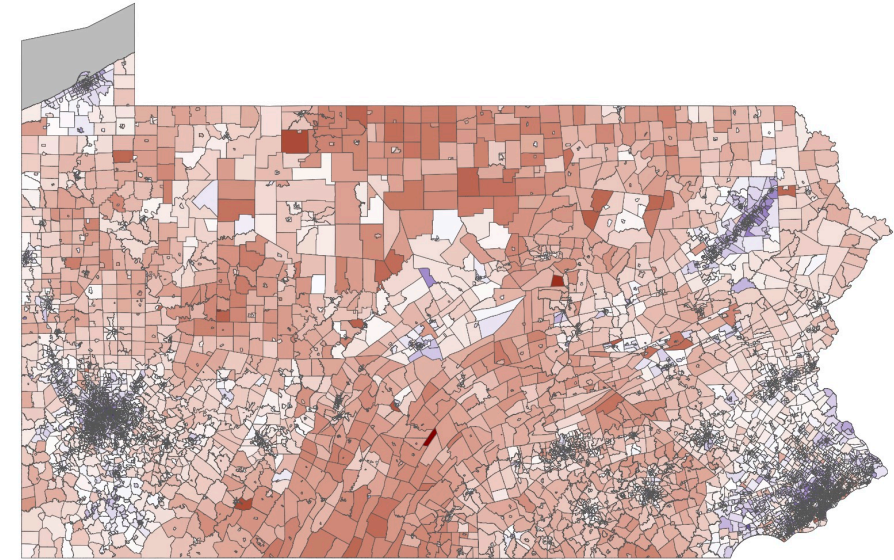
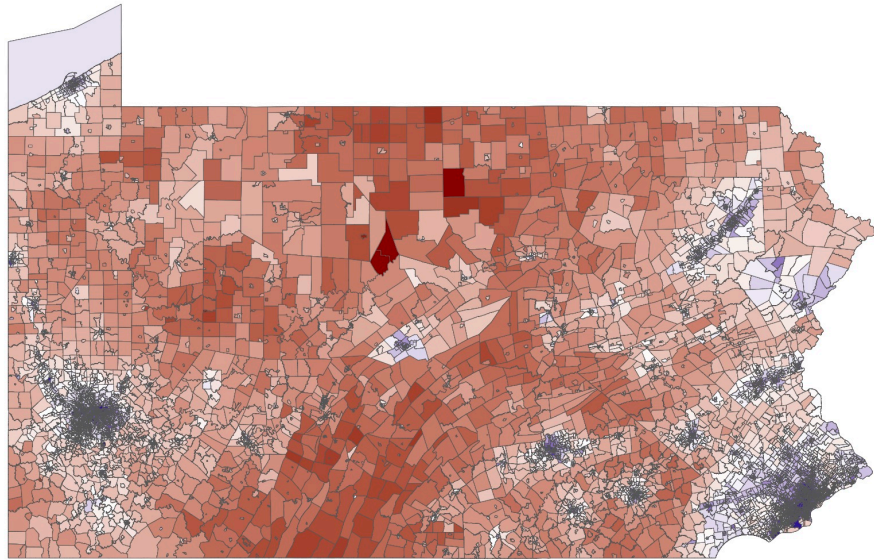
MONODIMENSIONAL POLITICAL SPACE

- ▶ DIME scores map candidates on a monodimensional political space (left-right spectrum)
- ▶ Therefore, we can run the following regression:

- ▶
$$\frac{2s_{Dg} - 1}{x_D - x_R} = \frac{\beta_0}{a} + \frac{w\bar{x}_g}{a} + \sum_{m=1}^M \frac{\tilde{\beta}_{1m}\bar{c}_{mg}}{a} + \epsilon_g$$

- ▶ We are now able to estimate relative ideological positions of precincts ($\frac{w\bar{x}_g}{a}$) and the effect of each covariate ($\frac{\tilde{\beta}_{1m}}{a}$)

ESTIMATES OF IDEOLOGICAL POSITIONS WITHIN PRECINCTS



- ▶ top-left: average Democratic share in each precinct in statewide elections from 2016 to 2020
- ▶ top-right: ideological estimate of each precinct accounting for ethnical composition of precincts
- ▶ bottom-left: ideological estimate of each precinct accounting for both ethnical composition and share of people with a college degree

LIMITATIONS OF THIS FIRST EXPERIMENT

- ▶ The estimates of both ideological positions of precincts and effects of demographic characteristics are not very interesting: we are mainly decomposing partisanship into ethnic and education features, trying to “water out” other differences
- ▶ That, however, is quite different from our original scope
- ▶ The main issue there is the monodimensional political space

ESTIMATING CANDIDATES' POSITIONS IN A MULTIDIMENSIONAL SPACE

- ▶ We are then interested in placing candidates in a multidimensional space: that also helps in understanding the effect of personal characteristics on voters' choices
- ▶ Various methodologies have been applied to this problem (voting scores, candidates' websites) but there is a lack of public datasets that cover both winners and losers
- ▶ Then, we tried to develop our own

A BAYESIAN ESTIMATE OF POSITIONS IN POLITICAL SPACE

- ▶ Using the model specification that we derived initially

$$\mathbb{E}[s_{Dg} - \frac{1}{2}] = (\mathbf{x}_D - \mathbf{x}_R)' \left(\frac{\beta_0}{a} + \mathbf{W} \frac{\bar{x}_g}{a} + \sum_{m=1}^M \frac{\tilde{\beta}_{1m}}{a} \bar{c}_{mg} \right) \text{ and the}$$

observed vote shares, we are now willing to estimate not only average ideological positions of people within a precinct, but also ideological positions of the candidates

- ▶ We just adopt a prior on distribution of precinct positions (based on what we estimated before) and a prior on ideological positions of the candidates on the first dimension (based on DIME)